ECDLP follow-ups: factor-base sweep, pencil pilot and Kummer gate

Reproducible experiment report

21 October 2026

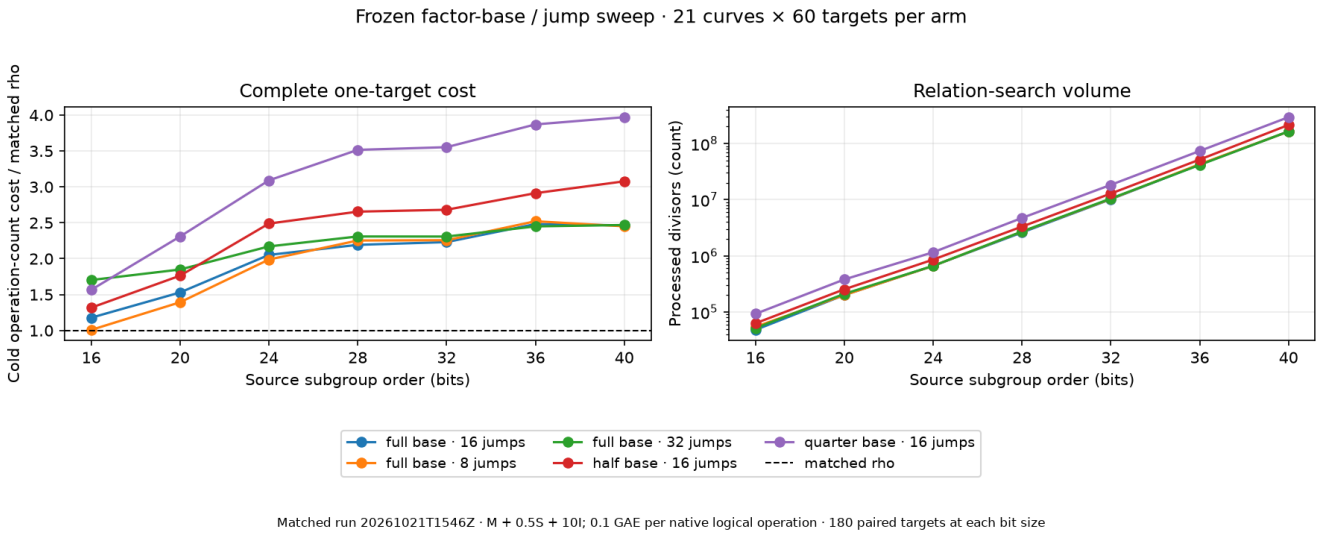
Scope. The frozen protocol covers four leads on 21 exact-order Scholten models, 60 separate targets per model. This report contains completed factor-base and jump experiments, a 420-target pencil pilot, checked branches of the published complete law, and an exact Kummer applicability gate. The full inversion-free law and full-cohort pencil collector remain open. All costs are cold one-target operation-count prices, not wall-speed claims.

Factor-base and jump sweep

Five arms solved and source-verified all 1,260 targets each, with zero failures or cap exhaustions. All 6,300 same-process rho controls were also source-verified. All 1,260 incumbent count and phase records replay exactly. At $M + 0.5S + 10I$ and 0.1 source-addition equivalents (GAE) per declared native logical operation, matched negation rho totals 1,088,802,370 GAE.

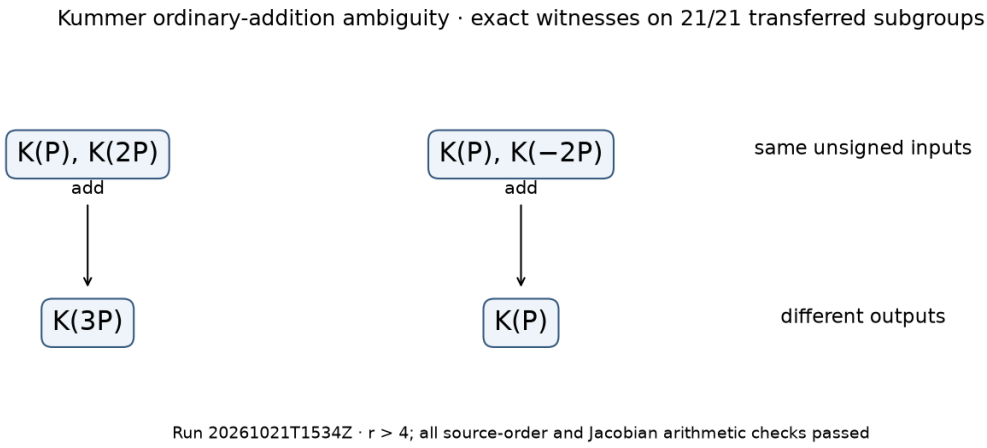
Arm	Divisors processed	Total GAE	/ external	/ rho
Full base, 16 jumps	219,581,773	2,666,291,894	1.000000	2.448830
Full base, 8 jumps	220,092,171	2,664,487,778	0.999323	2.447173
Full base, 32 jumps	219,864,588	2,671,135,032	1.001816	2.453278
Half base, 16 jumps	282,983,105	3,285,713,695	1.232316	3.017732
Quarter base, 16 jumps	392,583,830	4,267,226,320	1.600435	3.919193

The half and quarter bases rejected 35,385,164 and 110,402,348 split relations, respectively, when neither root was in the fixed target-blind base; all rejected candidates were charged. None of the five arms is below rho in the 36 pooled pricing cells. The 8-jump arm is below the 16-jump arm in 15 of 36 cells. Across bit sizes it is below at 16, 20, 24 and 40 bits, and above at 28, 32 and 36 bits. One seed was used per target; no uncertainty interval or isolated wall-time result was measured.



Kummer ordinary-addition gate

Set $P = \tau(G)$ and $Q = 2P$. The unsigned quotient gives $K(Q) = K(-Q)$, but $K(P+Q) = K(3P)$ and $K(P-Q) = K(P)$ differ for order $r > 4$. The source-order, Jacobian-arithmetic and distinct-output checks passed on 21/21 exact-order models. This rules out binary ordinary addition from two unsigned Kummer inputs for the current arbitrary-jump walk. A differential walk with a known difference and sign/lift recovery has not been implemented or priced.

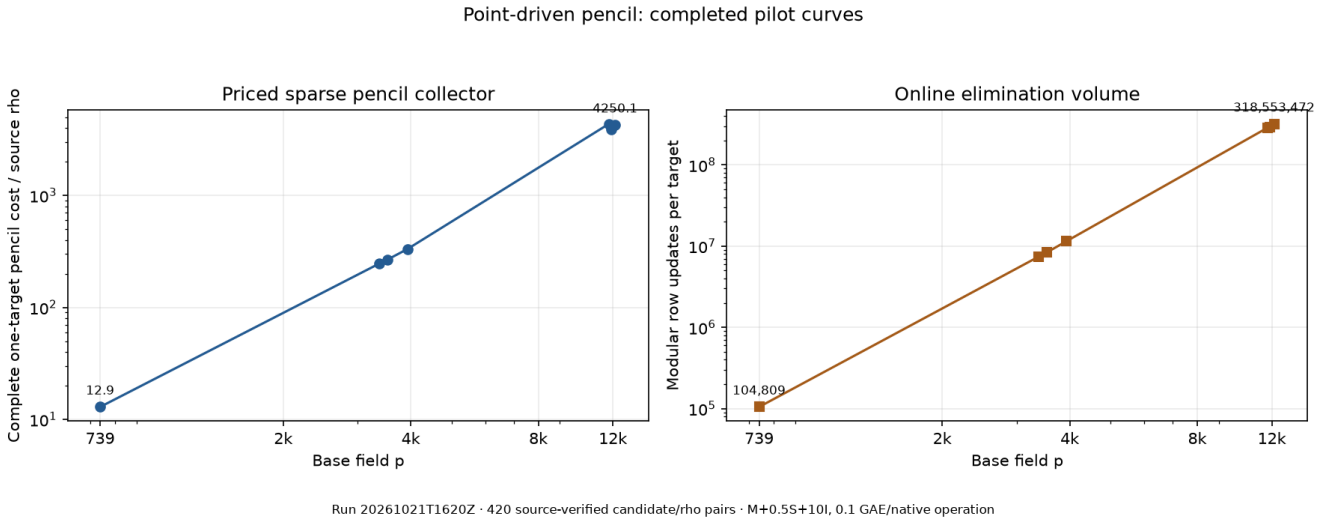


Point-driven pencil pilot

Seven original models, $p = 739, 3371, 3527, 3931, 11731, 11887, 12143$, each ran 60 separate targets. All 420 candidate and same-process rho scalars passed source multiplication. The independent validator recomputed all 420 scalar pairs; their replay matched every non-wall field in seven raw files. The direct F739 port also reproduced all 61 prior records after removing wall fields. Full measured setup cost is charged to each cold one-target solve.

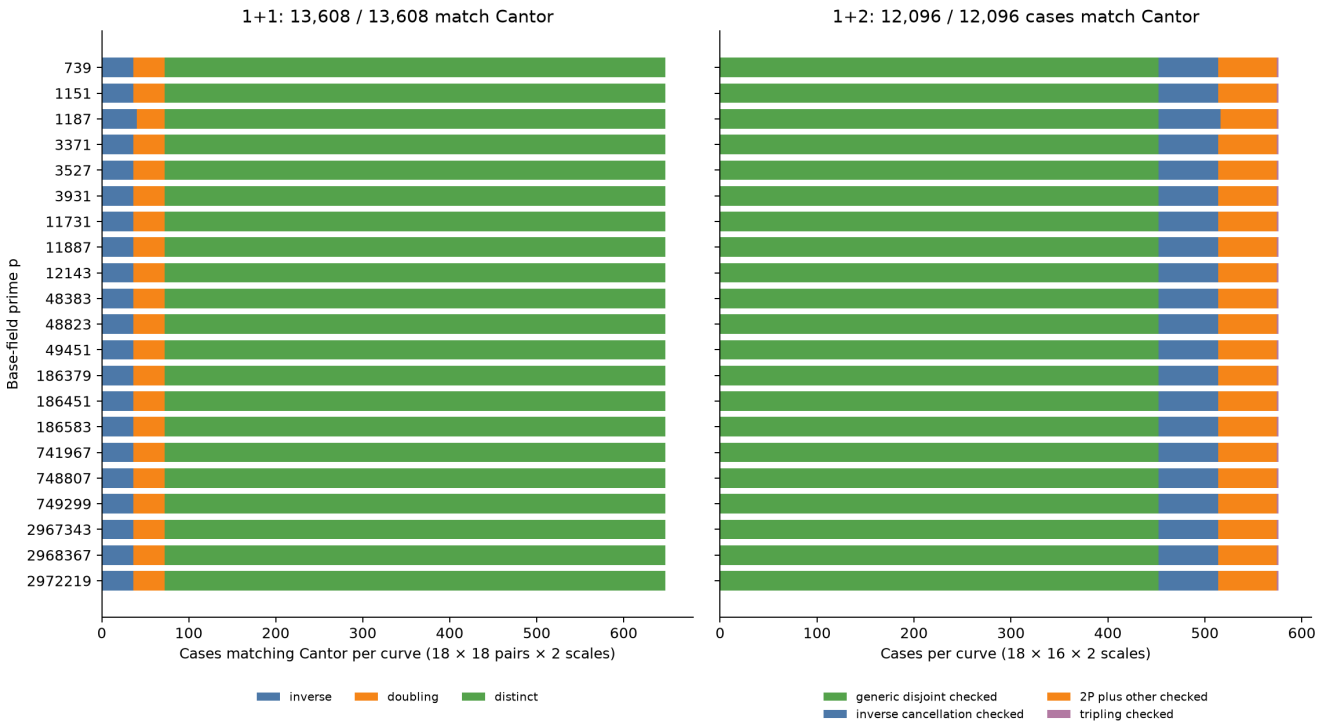
p	Verified	Relations	Modular row updates	Pencil/rho
739	60/60	21,550	6,288,519	12.931178
3,371	60/60	98,666	442,470,426	246.117110
3,527	60/60	103,256	499,795,794	267.374941
3,931	60/60	114,989	690,243,338	332.187349
11,731	60/60	343,641	17,235,887,587	4,331.361006
11,887	60/60	348,343	17,655,108,982	3,856.238613
12,143	60/60	355,615	19,113,208,311	4,250.094923

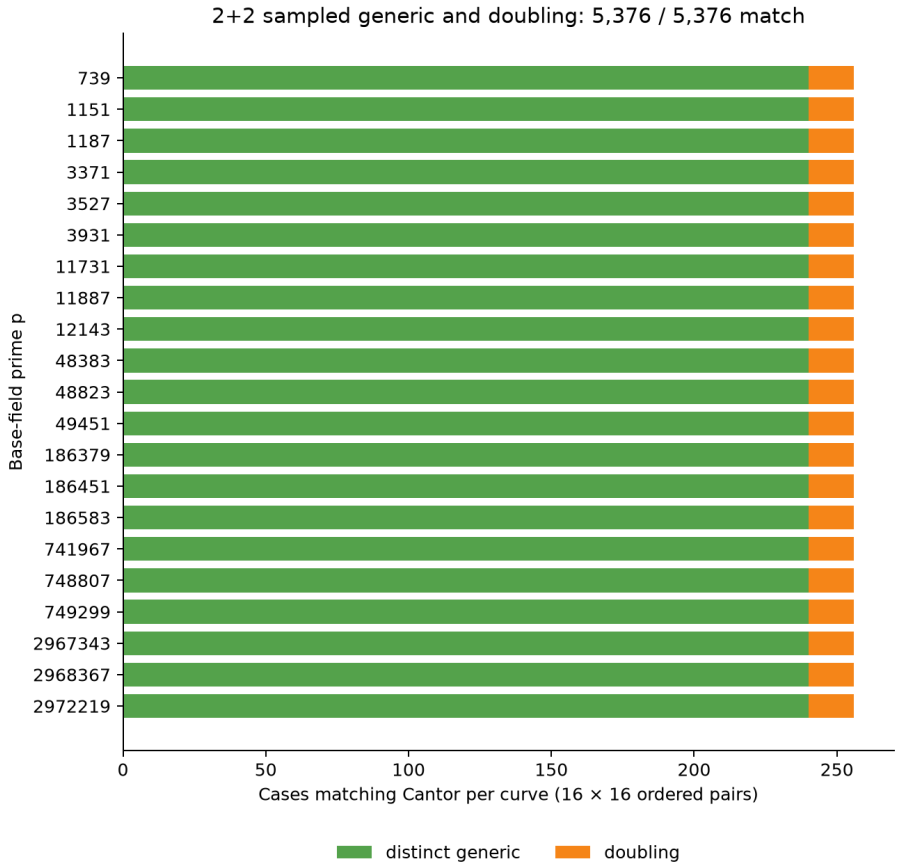
The price is $M + 0.5S + 10I$ plus 0.1 GAE per native logical operation; the independent validator recomputed 21 headline values from all seven models; the analyzer matched fresh rho counters to frozen controls on all seven. The 36-cell price grid and per-target JSONL are in `results/pencil-pilot/`. Upper-size runs are still open.



Complete-law branch gate

Equations (13) through (18) of https://doi.org/10.1007/s13389-024-00351-7 were implemented independently. Across the 21 normalized models and two projective scales, 13,608/13,608 degree-one-pair additions matched Cantor: 760 inverse pairs, 752 doublings and 12,096 distinct-point additions. All 12,096 sampled 1+2 cases matched Cantor: 9,492 generic disjoint, 1,304 inverse-support cancellations, 1,258 repeated-support additions with a distinct remaining point, and 42 triplings. Generic distinct 2+2 addition matched Cantor on 5,040 sampled cases and nonexceptional doubling on 336. Other 2+2 exceptional branches and the full DLP cohort remain unexecuted.





Open requirements and evidence

The published complete inversion-free genus-2 formulas are only partly implemented and have not run the 1,260 targets. The point-driven pencil collector has 420 new complete target solves but no full-cohort result here. A

differential Kummer collector remains unimplemented. These are open requirements, not negative end-to-end verdicts.

The machine record consists of 35 raw matched JSONL arm files, 6,300 paired candidate/rho target rows, the full 36-cell price grid, summary JSON, 21 Kummer witnesses, seven complete pencil JSONL files with a 36-cell-per-model price grid, and a checksum manifest under `research/four_followups_20261021/`. The native analyzer pairs workload hashes, seeds, caps and source scalars; the full-base arm reproduces the frozen external-law counts and phases. A separate native validator recomputed 15 headline quantities from raw rows; an archive-input rerun was compared byte-for-byte across all 35 files. The plotting script only renders those recorded values. See https://github.com/aburan28/crypto/pull/2304 for the review branch and https://eprint.iacr.org/2024/623.pdf for the separate complete-law path.